MILITARY SPECIFICATION

ZIRCONIUM CARBIDE, POWDERED, TECHNICAL

This specification has been approved by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the procurement of one type of low hafnium content zirconium carbide hereinafter referred to as the material.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

WS 4503B Rocket Motor, Mk 56 Mods 0, 1 & 2

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (Code Ident 10001)

OD 18871 Volatiles by Drying in an Oven or Furnace, Determination of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6810

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:21Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 329	Apparent Density of Refractory Metals and Compounds by the Scott Volumeter
ASTM B 330	Average Particle Size of Powders of Refractory Metals and Compounds by the Fisher Sub-Sieve Sizer
ASTM E 146	Chemical Analysis of Zirconium and Zirconium-Base Alloys

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS.

3.1 Preproduction sample. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of the material is required and shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 for the purpose of determining that, prior to starting production, the contractor's production methods are capable of producing material that complies with the technical requirements of the contract (see 4.5).

3.2 Chemical and physical properties. The chemical and physical properties of the material shall be in accordance with TABLE I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:21Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical and physical properties.

Characteristic	Minimum	Maximum
Composition		
Zirconium (Zr plus Hf), wt%	87.4	88.5
Hafnium (Hf), wt%	—	2.0
Total carbon (C), wt%	10.8	11.9
Free carbon (C), wt%	—	1.5
Iron (Fe), wt%	—	0.05
Oxygen (O), wt%	—	0.6
Nitrogen (N), wt%	—	0.8
Properties		
Apparent density, g/cm^3	1.46	1.83
Average particle diameter, µm	1.5	3.0
Volatiles content, wt%	—	0.10

3.3 Product and process changes. The contractor shall notify the procuring activity of any changes in the product or processing techniques.

3.3.1 Mark 56 Mod 2 Dual Thrust Rocket Motor procurement. When the zirconium carbide is procured for the Mk 56 Mod 2 Dual Thrust Rocket Motor, the requirements for change of material, material vendors, processes or equipment shall be those in WS 4503 paragraph 3.1.3 (6.2.1).

3.4 Workmanship. The material shall be uniform, free from contamination, foreign material or any other defect that would prevent its use for the purpose intended. The material shall be manufactured in accordance with standard manufacturing procedures of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.5)

b. Quality conformance inspection (see 4.6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:21Z
Check the source to verify that this is the current version before use.

4.4 Certificate of compliance. When specified in the contract (see 6.2), the contractor shall furnish a certificate of compliance stating that the material is in conformance with this specification.

4.5 Preproduction inspection. Unless otherwise specified in the contract (see 6.2), a preproduction sample consisting of sufficient material to perform the tests of 4.7 shall be manufactured in accordance with 3.1 and forwarded to an activity designated by the procuring activity (see 6.2) for preproduction inspection. The preproduction inspection shall consist of tests and examinations specified in TABLE II. Acceptance of the preproduction sample shall be based on no defects in the sample. Any production of the material by the contractor, prior to approval of the preproduction sample, shall be at the contractor's risk.

4.6 Quality conformance inspection. The quality conformance inspection shall consist of the tests and examinations specified in TABLE II performed on samples selected in accordance with 4.6.2. If any sample does not conform to any requirement of this specification, the lot shall be rejected.

4.6.1 Lot size. A lot shall consist of all material, produced under essentially the same conditions, by the same process, by one manufacturer, at one plant, and to be offered for acceptance at one time.

4.6.2 Quality conformance inspection sampling. Four containers selected at random from each lot shall be individually sampled. If the lot consists of less than four containers, each container shall be sampled. If the lot consists of only one container, the container shall be sampled at two locations.

TABLE II. Preproduction inspection and quality conformance inspection.

Tests and Inspections	Requirements	Preproduction Inspection	Quality Conformance Inspection	Test Methods
Packaging and marking	Section 5		X	Visual
Zirconium	Table I	X	X	4.7.1
Hafnium	Table I	X	X	4.7.2
Total carbon	Table I	X	X	4.7.3
Free carbon	Table I	X	X	4.7.4
Iron	Table I	X	X	4.7.5
Oxygen	Table I	X	X	4.7.6
Nitrogen	Table I	X	X	4.7.7
Density	Table I	X	X	4.7.8
Avg particle diameter	Table I	X	X	4.7.9
Volatiles content	Table I	X	X	4.7.10
Workmanship	3.4	X	X	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:21Z
Check the source to verify that this is the current version before use.

4.7 Tests.

4.7.1 Zirconium. The zirconium content shall be determined by difference (i.e. sum of the measured elements except hafnium subtracted from the total).

4.7.2 Hafnium. The hafnium content shall be determined by X-ray fluorescence or equivelent.

a. Conditions

Analytical line	Hf 1 $L\beta_1$
Background	41.00
Crystal	LiF 200
Medium	Vacuum
Count time	40 sec
Anode material	Cr
KV	50
Ma	30
Detector	Scintillation
Collimator	Fine
Vessel	Steel + carbon disk

b. Sample preparation

Borax - beads

(1) containing 5 grams(g)$Li_2B_4O_7$ + 0.2 g sample

(2) containing 5 grams(g)$Li_2B_4O_7$ + 0.2 g sample plus known amounts of pure HfO_2 (1%, 2%, 5%)

4.7.3 Total carbon. The total carbon shall be determined by the micro-Pregl combustion technique or equivalent.

4.7.4 Free carbon. The free carbon shall be determined as follows:

a. Approximately 1 to 2 g of sample are weighed into a platinum crucible. Add 25 cm^3 of concentrated hydrofluoric acid and then heat on a sand bath. Nitric acid is added dropwise until sample is dissolved. Distilled water is added and the contents of the crucible are quantitatively rinsed into a porcelain filter crucible. Filter crucible and contents are then dried in an oven at 105°C.

b. The filtration residue is carefully scraped off with a sharp spatula (the bottom of the crucible has to be completely white if this is properly performed), onto a porcelain combustion boat and the carbon content determined by the micro-Pregl combustion technique or equivalent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:21Z
Check the source to verify that this is the current version before use.

4.7.5 Iron. The iron content shall be determined in accordance with ASTM E 146 using a 1.000 g sample that is first calcined in air. Dissolve the oxide in about 20 cm^3 of hydrofluoric acid by gently heating. After complete dissolving, cautiously add 10 cm^3 of sulfuric acid and boil the solution to dense white fumes. Continue as described in the ASTM method. An equivalent method may be used.

4.7.6 Oxygen. The oxygen content shall be determined by the micro Unterzaucher combustion technique or equivalent.

4.7.7 Nitrogen. The nitrogen content shall be determined by the Dumas method or equivalent.

4.7.8 Density. The density shall be determined in accordance with ASTM B 329.

4.7.9 Average particle diameter. The average particle diameter shall be determined in accordance with ASTM B 330 using 6.73 sample.

4.7.10 Volatiles content. The volatiles shall be determined in accordance with OD 18871 using an appropriate size sample dried for 2 hours at 105 $\pm$ 2°C.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract (see 6.2), packaging and packing shall be level C.

5.1.1 Level C. Unless otherwise specified in the contract (see 6.2), packaging and packing of the zirconium carbide shall be in accordance with standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination. Container size shall be as specified in the contract (see 6.2). Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129. Marking shall include but not be limited to, the following information.

a. Title, number and date of this specification

b. Manufacturer's name and location

c. Material trade name.

d. Net weight and volume.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:21Z
Check the source to verify that this is the current version before use.

e. Lot number, batch number(s), and date of manufacture.

f. Storage conditions.

g. Toxicity precautions, if applicable.

h. Contract number.

6. NOTES

6.1 Intended use. Zirconium carbide powder is intended to be used as a propellant ingredient for the Mark 56 Mod 2 Dual Thrust Rocket Motor

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Quantity required.

c. Place of delivery.

d. When a preproduction sample is not required (see 3.1, 4.5 and 6.4).

e. Inspection conditions when other than as specified (see 4.2).

f. Assigned activity for preproduction inspection (see 4.5).

g. Packaging requirements if other than as specified (see 5.1).

h. Size of container required (see 5.1.1).

i. Any special marking required (see 5.2).

j. Changes to material, material vendors, processes and equipment (see 3.3.1).

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data quirement	Applicable DID*
4.4	Certificate of compliance	DI-E-2121

*DIDs (Data Item Description/DD Form 1664) for the above requirements are documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:21Z
Check the source to verify that this is the current version before use.

6.3 Rounding-off specified limits. The following applies to all specified limits in this specification. For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29, Standard Recommended Practice for Indicating Which Places of Figures are to be Considered Significant in Specified Limiting Values.

6.4 Preproduction sample waiver. Preproduction samples submitted and approved on a recent contract may be accepted by the procuring activity in lieu of an additional preproduction sample inspection. When the preproduction sample is waived (see 6.2.1d), the procurement document should contain a statement specifying that the standards of workmanship exhibited by the previously approved preproduction sample shall determine the minimum requirements of the current contract or order.

6.5 Suggested source of supply. A product that has met the requirements of this specification in past procurement actions is manufactured by Herman C. Starck, Inc., 210 Park Ave., West Building 24th Floor, New York, NY 10017. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

Custodian:
NAVY - OS

Preparing Activity:
NAVY - OS

Project Number
6810-NA 47

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:21Z
Check the source to verify that this is the current version before use.